

How Much of the Delayed-Feedback Penalty Is Recoverable?

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Abstract. Conversion labels arrive late, so a model trained today must either wait for labels and train on stale data, or use fresh data whose labels are only partly observed. We derive a closed-form ceiling on how much of the resulting loss a censoring-corrected estimator can recover: it is the mean relative Fisher information of the censored observations, computable from the empirical delay distribution alone, before any model is trained. On the Criteo delayed-feedback corpus that ceiling is 0.868 for a seven-day attribution window. We test it with 1,600 runs spanning five model capacities, twenty rolling origins and five delayed-label strategies under one protocol. Censoring correction attains 0.883, 0.874 and 0.849 of the penalty at the three smallest capacities, within 0.02 of the predicted ceiling, then falls to 0.625 at the largest. We show the shortfall is misspecification rather than estimation error: the delay distribution varies sharply with the covariates, and only a model expressive enough to resolve that variation is harmed by assuming it away. We further show that the apparent advantage of knowledge distillation here is mostly capacity transfer rather than delay mitigation, and give the control that separates the two.

Keywords: Delayed feedback · Conversion rate prediction · Censored labels · Knowledge distillation · Computational advertising

1 Introduction

[To be written. Contribution bullets to land here: (i) a parameter-free ceiling on recoverable delayed-feedback loss, validated to within 0.02; (ii) the capacity regime in which censoring correction saturates that ceiling and the regime in which it does not, with the measured cause; (iii) a control separating distillation’s capacity transfer from genuine delay mitigation.]

2 Problem Setting

A click occurs at time t with covariates x . If it converts, the conversion follows after a delay $d > 0$; write $d = \infty$ for a click that never converts. Under an attribution window Δ the prediction target is

$$y = \mathbf{1}[d \leq \Delta], \quad \eta(x) = \Pr(d \leq \Delta \mid x). \quad (1)$$

Table 1. The five strategies, with the infeasible oracle and the distillation control. All are evaluated on the same held-out day against the same target (1)

	training set	target
Oracle	$\mathcal{M} \cup \mathcal{C}$	true y (infeasible at T)
A, wait	$\mathcal{M}$	y
B, distil	$\mathcal{M} \cup \mathcal{C}$	teacher’s $\hat{\eta}$; teacher trained on $\mathcal{M}$
B [−] , control	$\mathcal{M}$	the same teacher’s $\hat{\eta}$
C, correct	$\mathcal{M} \cup \mathcal{C}$	y_{obs} ; loss on $\sigma(z)\hat{F}(e)$, serve $\sigma(z)$
D, band	$[0, T - u_k), k = 1 \dots K$	$\mathbf{1}[u_{k-1} < d \leq u_k]$, predictions summed
E, reweight	$\mathcal{M}$	y , weighted by $\hat{p}_{\mathcal{C}}(x)/\hat{p}_{\mathcal{M}}(x)$

The difficulty is not that y is noisy but that it is not yet determined. At an origin T , the moment a training set is assembled, a row clicked at t has been observable for

$$e(t) = \text{clip}(T - t, 0, \Delta), \quad y_{\text{obs}} = \mathbf{1}[d \leq e(t)] \leq y. \quad (2)$$

Two regimes follow. If $t \leq T - \Delta$ then $e = \Delta$ and $y_{\text{obs}} = y$ exactly. If $t > T - \Delta$ the row is right-censored: $y_{\text{obs}} = 1$ is conclusive, but $y_{\text{obs}} = 0$ is ambiguous, because the click may still convert inside its remaining window. Everything below follows from (2), and no strategy is granted an assumption the others lack.

2.1 Five Strategies as Estimators

Let $\mathcal{M} = [0, T - \Delta)$ denote the mature rows and $\mathcal{C} = [T - \Delta, T)$ the censored ones, with $|\mathcal{M}| = m$ and $|\mathcal{C}| = c$. Table 1 states each strategy as a choice of training set and target. The oracle is infeasible at time T , since it uses labels that do not yet exist; it serves only as the denominator for how much of the penalty a feasible strategy recovers.

Strategy C is the classical joint conversion-delay likelihood [1] with a non-parametric $\hat{F}$. Under the conditional-independence assumption

$$\Pr(d \leq e \mid x, d \leq \Delta) = F(e), \quad (3)$$

that is, delay independent of the covariates given conversion inside the window, the observed label has mean $\Pr(y_{\text{obs}} = 1 \mid x, e) = \eta(x)F(e)$. Fitting $\sigma(z(x))\hat{F}(e)$ against y_{obs} under log loss is then a proper composite likelihood for η . The correction lives in the loss, and $\sigma(z)$ is served directly with no post-hoc rescaling.

B[−] is not an ablation but a necessary control. Strategy B changes two things at once relative to A: the student inherits a teacher that may be larger than itself, and it trains on covariates that A had to discard. B[−] removes only the second. Without it, any gain from B is unattributable.

3 How Much Is Recoverable?

We now bound what any estimator of the form C can recover, using only the delay distribution.

Under (3) a censored observation with window e is Bernoulli with mean $\eta F(e)$. Its Fisher information about η is $I(\eta; e) = (\partial_\eta \eta F)^2 / (\eta F(1 - \eta F)) = F / (\eta(1 - \eta F))$, against $I_{\text{full}} = 1 / (\eta(1 - \eta))$ for a mature row. The relative information of a censored row is therefore

$$\rho(F) = \frac{I(\eta; e)}{I_{\text{full}}} = \frac{F(1 - \eta)}{1 - \eta F} \in [0, 1], \quad \rho(1) = 1, \quad \rho(F) \approx F \text{ for small } \eta. \quad (4)$$

Proposition 1. *Assume (3), and that excess risk scales as $1/n$ in the effective sample size. Let $\bar{\rho} = |\mathcal{C}|^{-1} \sum_{i \in \mathcal{C}} \rho(F(e_i))$. Then the fraction of the oracle-minus-wait gap recovered by strategy C is*

$$R = \frac{1/m - 1/(m + \bar{\rho}c)}{1/m - 1/(m + c)} = \frac{\bar{\rho}(m + c)}{m + \bar{\rho}c}. \quad (5)$$

R contains no fitted quantity: $\bar{\rho}$ follows from the empirical delay distribution and the base rate, and m and c from the calendar. It can be computed before training anything.

On this corpus the delay distribution is strongly front-loaded, with 57.4% of conversions within seven days landing in the first hour, so censored rows retain most of their information even when their window has barely opened. Averaged over the twenty origins used below, $\bar{\rho} = 0.886$ and $R = 0.888$ for $\Delta = 1$ d, and $\bar{\rho} = 0.847$ and $R = 0.868$ for $\Delta = 7$ d.

Proposition 1 prices the statistical cost of censoring only. It assumes (3) holds, and it attributes the whole oracle-minus-wait gap to effective sample size, ignoring covariate shift between $\mathcal{M}$ and the served day. Section 5 shows both idealisations are visible in the data, and identifies which one binds.

4 Experimental Design

Protocol. We use rolling origins $T = 40, \dots, 59$, each with training windows ending at T minus that strategy’s own wait, and evaluation on the single day $[(T+1)\text{d}, (T+2)\text{d})$ with one idle day in between, held fixed as the window slides. The label (1) is enforced identically in training and evaluation: a row may enter a training set only if its label is knowable at T . Evaluation days carry 230k to 300k rows. Conversions are logged to day 90.955 while clicks stop at day 61.001, so every evaluation label is fully resolved at every Δ . The origin is the unit of replication and all comparisons are paired on it, because the evaluation positive rate swings from 0.1595 to 0.1966 across these twenty days, an order of magnitude more than any effect we measure.

Table 2. The capacity ladder. Table parameters vary over $3,641\times$ and the resulting normalised entropy is approximately equally spaced

	S_0	S_1	S_2	S_3	S_4
hash rows	36	64	160	896	131,072
table parameters	2,592	4,608	11,520	64,512	9,437,184
NE ($\Delta=7d$, one origin)	0.78626	0.77312	0.75730	0.74124	0.72794

Tasks. We use $\Delta \in \{1d, 7d\}$ and drop $\Delta = 1h$, whose oracle gap is -0.00002 ($p = 0.842$): there is no penalty to recover, so every strategy ties with A by construction.

Capacity Ladder. Strategy rankings that hold at one model size need not hold at another, so the entire comparison is repeated at five capacities $S_0, \dots, S_4$. All five share one backbone and differ only in the number of categorical hash-table rows, a single monotone knob that is exactly the embedding-table memory footprint, which is the binding constraint in production ranking. The five are approximately equally spaced in normalised entropy by construction (mean step 0.0149; Table 2) and are monotone in it at 40 of 40 (task, origin) cells.

Teacher. Strategy B always uses the same teacher, selected as the best of a 21-rung architecture ladder and held at full table resolution. The teacher sees A’s window and A’s labels, so it differs from A on capacity alone, receiving no extra data and no extra label information.

Metrics. We report normalised entropy $NE = \text{logloss}/H(\bar{y})$, the calibration ratio $\bar{p}/\bar{y}$, and a calibration-adjusted NE obtained by multiplying every prediction by $c = \bar{y}/\bar{p}$, so that calibration is exactly one, and recomputing NE. That quantity can exceed NE, because matching the mean is not the same as minimising log loss; it is a level-matching diagnostic, not a lower bound. All models train for exactly one epoch: a second epoch already overfits at every capacity (train -0.0186 , evaluation $+0.0019$, $p = 0.025$).

Scale and Verification. The grid is $2 \text{ tasks} \times 5 \text{ capacities} \times 20 \text{ origins} \times 8 \text{ arms}$, or 1,600 scored runs. Before launching we verified that the calibration multiplier forces calibration to one at every starting bias; that the harness reproduces previously logged cells bit-exactly (14 of 14, absolute difference 0); that the shared teacher is identical across capacities to seven decimal places while student models are not; and that every arm at a given (task, origin) is scored on an identical evaluation set.

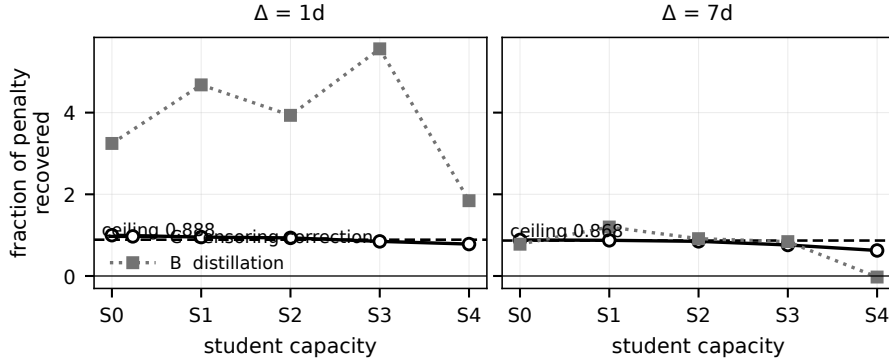


Fig. 1. Measured recovery of the delay penalty against the parameter-free ceiling of Proposition 1 (dashed). Censoring correction (C, open circles) tracks the ceiling at the three smallest capacities and falls away at the two largest. Distillation (B, grey squares) behaves differently and exceeds one at S_1 for $\Delta = 7d$, because the oracle is capped at the student’s capacity whereas the teacher is not

5 Results

5.1 The Penalty Depends on the Window, Not Much on Capacity

The oracle-minus-wait gap at $\Delta = 7d$ is $+0.00538, +0.00602, +0.00746, +0.00842$ and $+0.00598$ for S_0 through S_4 , all with $p < 0.001$ over 20 of 20 origins. At $\Delta = 1d$ it is an order of magnitude smaller, from $+0.00067$ to $+0.00092$. Waiting is expensive only when the window is long relative to the available history: at seven days it withholds 12.6 to 18.6% of the corpus.

5.2 The Ceiling Is Attained, and Then It Is Not

Figure 1 is the central result. For $\Delta = 7d$ the predicted ceiling is 0.868, and C measures 0.883, 0.874, 0.849, 0.759 and 0.625 across S_0 through S_4 . At the three smallest capacities the prediction is accurate to $+0.015, +0.006$ and -0.019 , from a formula with no free parameters that was fitted to nothing. At S_3 and S_4 it over-predicts by 0.11 and 0.24.

5.3 The Shortfall Is Misspecification, Not Estimation

Proposition 1 assumes (3), and that assumption is false on this corpus in a way we can measure. Stratifying converted rows by the distinct values of a single categorical field, the share of within-seven-day conversions landing in the first hour ranges from 0.453 to 0.689, and the median delay from 0.16 h to 1.73 h, a tenfold spread (Table 3).

This resolves the pattern in Fig. 1. A model too small to resolve covariate-dependent delay cannot be harmed by a correction that assumes it away, so at

Table 3. The delay distribution is strongly covariate-dependent, violating (3). Strata are distinct values of one categorical field; the pooled $F(1h)$ is 0.574

stratum	1	2	3	4	5	6
conversions $\leq 7d$	860,851	441,893	434,540	192,949	134,479	23,790
$F(1h)$	0.453	0.576	0.602	0.628	0.689	0.687
median delay (h)	1.73	0.41	0.29	0.34	0.23	0.23

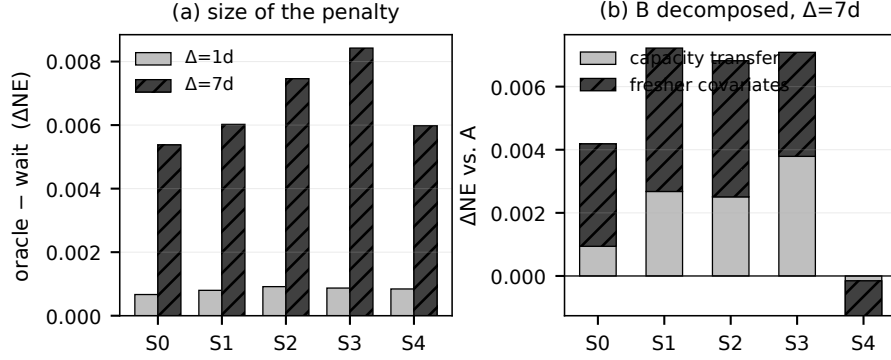


Fig. 2. (a) The delay penalty by capacity and window. (b) Strategy B decomposed using the B^- control into capacity transfer and exposure to fresher covariates. At $\Delta = 7d$ the fresh-covariate term is 47 to 78 % of the gain; at $\Delta = 1d$, where little penalty exists to recover, it is 6 to 29 %

S_0 through S_2 the only cost of censoring is the statistical one that Proposition 1 prices exactly. As capacity grows, the misspecification bias in $\hat{F}(e)$ becomes expressible and recovery falls below the ceiling. The prediction is therefore best read as an upper bound that is tight precisely when the model is sample-limited rather than bias-limited, and the gap between prediction and measurement estimates what a covariate-conditional delay model would buy.

5.4 Distillation’s Advantage Is Mostly Not About Delay

Strategy B beats A at $\Delta = 7d$ by +0.00419 to +0.00722 for S_0 through S_3 , all with $p \leq 0.001$ over 19 of 20 origins. Read alone this looks like delay mitigation. The B^- control shows it is largely not: capacity transfer accounts for 22 to 53 % of the gain at $\Delta = 7d$ across S_0 through S_3 , and 71 to 94 % at $\Delta = 1d$ (Fig. 2b). Only the residual, the student’s exposure to covariates that A discarded and which carry no new label information, is attributable to the delayed label; it is significant at every capacity ($p \leq 0.001$) but never dominant at the short window. At S_4 , where the teacher’s headroom over the student is only +0.0036, B collapses to -0.00015 ($p = 0.898$): distillation helps exactly while the teacher

is meaningfully better than the student, which is a statement about model size rather than about feedback delay.

Strategies D and E do not repay their complexity. D ranges from -0.00215 to $+0.00457$ with no monotone pattern in capacity; its slowest band still trains on A’s window and so never escapes the wait it was designed to avoid. E stays within ± 0.0027 of A everywhere, consistent with the penalty being missing labels rather than covariate staleness, and E by construction never touches an unresolved label.

6 Related Work

[To be written. Anchor on the joint conversion-delay likelihood [1]; the importance-weighting line [4, 7, 5]; stream-level alternatives [3, 2]; and covariate-conditional delay [6, 8], which targets exactly the assumption Section 5 shows is violated.]

7 Limitations

We use one seed per origin. The origin is the unit of replication ($n = 20$), and seed sensitivity was characterised separately at a standard deviation of 0.0004 to 0.0030, an order of magnitude below the $\Delta = 7$ d effects, but seeds are not averaged here. The twenty origins share training data heavily, so the effective sample size is below twenty. Proposition 1 assumes a $1/n$ excess-risk rate, which we do not verify directly; its empirical accuracy at S_0 through S_2 is the evidence offered. The calibration multiplier is fitted on evaluation labels and is a diagnostic, never an achievable score. Strategy B’s soft targets are largely teacher in-sample on $\mathcal{M}$, and out-of-fold targets, the standard remedy, are untested. Finally, all results come from a single public corpus with one delay distribution; the ceiling is corpus-specific even though the formula is not.

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